

SIMATSENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55%

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I Kamalli shree V(192525030)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent(Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5– Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5– Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5– Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5– Insufficient testing and evaluation.

CO1:ASSESSMENTTOOL1

Name:KamallishreeV Reg.

No.: 192525030

Department:AI&ML

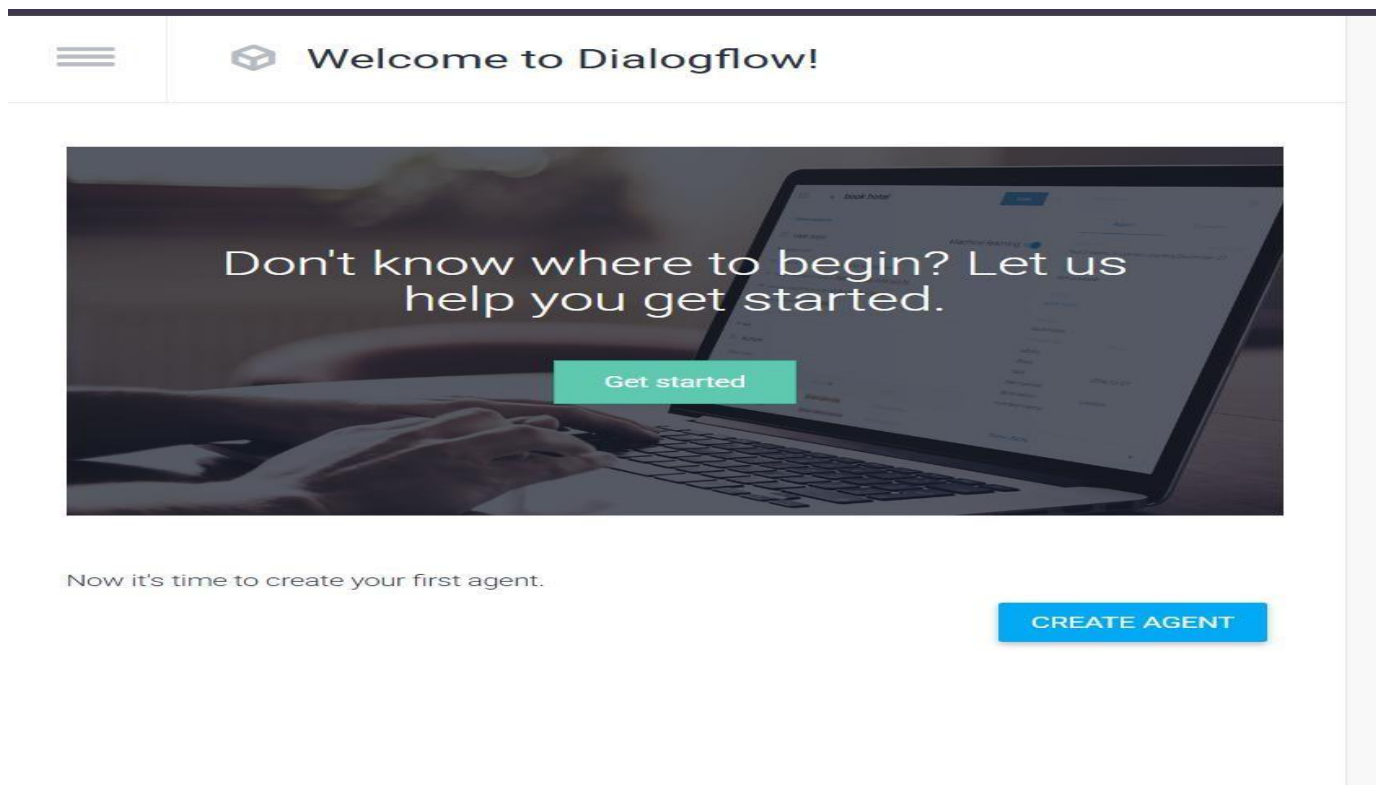
Course:MLA0103-Artificialintelligenceandexpertsystems

Problem Analysis

Content:

The university receives a large number of student queries every day regarding course registration, examination schedules, attendance details, timetables, and campus services. Handling these queries manually requires significant time and effort from administrative staff, leading to delays in responding to students.

An AI-based Student Support Assistant can solve this problem by providing instant and accurate responses to frequently asked questions. Using Natural Language Processing (NLP), the chatbot understands user queries, identifies the correct intent, and returns the appropriate response. This reduces the workload of university staff, improves student satisfaction, and ensures that assistance is available 24 hours a day.



AI Concept Identification

Add this paragraph

The proposed chatbot uses Artificial Intelligence techniques to understand and process student queries. Natural Language Processing enables the chatbot to recognize user language, while Dialogflow classifies the user's intent and extracts important keywords using entities. Based on the identified intent, the chatbot retrieves the appropriate predefined response. If the system cannot recognize the query, it activates the Default Fallback Intent and requests the user to rephrase the question.



StudentSupportAssistant

CREATE

DEFAULT LANGUAGE

English — en



Primary language for your agent. Other languages can be added later.

DEFAULT TIME ZONE

(GMT+6:00) Asia/Almaty



Date and time requests are resolved using this timezone if not provided in the API requests.

GOOGLE PROJECT

Create a new Google project



Enables Cloud functions, Actions on Google and permissions management.

AGENT TYPE



Set as Mega Agent

Combine multiple Dialogflow agents (i.e. sub agents) into a single agent (i.e. [mega agent](#)).



Intents

[CREATE INTENT](#)

Default Fallback Intent



Default Welcome Intent

Add follow-up intent



No regular intents yet. [Create the first one.](#)

Intents are mappings between a user's queries and actions fulfilled by your software. [Read more here.](#)

Before you start, check out [Prebuilt Agents](#), a collection of agents developed by the Dialogflow team.

Course Registration Intent

The Course Registration intent is designed to answer student queries related to semester registration. When the student enters phrases such as "I want to enroll," "Registration process," "Open registration," or "How can I register?", Dialogflow identifies the Course Registration intent and returns information about the registration process.

The chatbot guides students to access the Student Portal and complete the registration before the deadline.



• Course Registration

SAVE

Contexts ?

Events ?

Training phrases ?

 Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.



Train the intent with what your users will say

Provide examples of how users will express their intent in natural language. Adding numerous phrases with different variations and parameters will improve the accuracy of intent matching. [Learn more](#)

ADD TRAINING PHRASES

Training phrases

Search training phrases 



Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

” I want to enroll

” Open registration



” Registration process

” Register for semester

” Course registration

” How can I register?

Responses



DEFAULT +

Text Response



- | | |
|---|---|
| 1 | Course registration opens at the beginning of every semester. Please login to the Student Portal and complete the registration before the deadline. |
| 2 | Enter a text response variant |

ADD RESPONSES

Intent saved

OK

Exam Schedule Intent

This intent provides examination-related information to students. Whenever a user asks about exam dates, semester examinations, or examination timetables, Dialogflow recognizes the intent and responds with the examination schedule available on the university website. This reduces unnecessary visits to the examination office.



Exam Schedule

SAVE

Training phrases ?

Search training phrases

 Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

” Add user expression

” Upcoming exams

” Exam timetable

” Semester exams

” When is my exam?

” Exam dates

Responses ?

^

DEFAULT +

Text Response ? 

1 The examination schedule is available on the university website under the Examination section.

2 Enter a text response variant

ADD RESPONSES

Attendance Intent

The Attendance intent allows students to obtain information regarding their attendance percentage. The chatbot recognizes queries such as "**Check attendance**," "**Attendance details**," and "**My attendance**" and directs students to view their attendance through the Student Portal.

The screenshot displays the Dialogflow console interface for configuring an intent named "Attendance". At the top, there is a blue "SAVE" button. Below the header, there are tabs for "Contexts", "Events", and "Training phrases". The "Training phrases" tab is active, showing a list of phrases: "Add user expression", "check attendance", "attendance details", "My attendance", and "Attendance percentage". A yellow warning banner states: "Template phrases are deprecated and will be ignored in training time. More details [here](#)." Below this, a text box explains: "When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types." At the bottom, there is a "Text Response" section with two variants: "1 You can view your attendance percentage through the Student Portal." and "2 Enter a text response variant". Below the responses, there is an "ADD RESPONSES" button and a toggle switch for "Set this intent as end of conversation".

Attendance SAVE

Contexts ? ▼

Events ? ▼

Training phrases ? Search training phrases Q ^

Warning: Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

- ☞ Add user expression
- ☞ check attendance
- ☞ attendance details
- ☞ My attendance
- ☞ Attendance percentage

Text Response 🔗 📄

1 You can view your attendance percentage through the Student Portal.

2 Enter a text response variant

ADD RESPONSES

☐ Set this intent as end of conversation ?

Timetable Intent

The Timetable intent provides students with information about their academic schedule. When students ask questions such as "**Show classes**," "**Today's timetable**," or "**Class schedule**," the chatbot directs them to the Academic Portal where the timetable is available.



Timetable

SAVE

Training phrases 

Search training phrases  

 Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

 Add user expression

 show classes

 time table

 class schedule

 **Today's** timetable

Campus Services Intent

The Campus Services intent provides information regarding university facilities such as the Library, Hostel, Transport, Medical Centre, and Student Support Office.

This intent improves accessibility by helping students quickly locate important campus services without contacting administrative staff.



Campus Services

SAVE



Training phrases 

Search training phrases  

 Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

” Add user expression

” wifi support

” medical centre

” transport

” hotel office

” library things

Entity Design

The chatbot uses a custom entity called **@services** to recognize different campus facilities. The entity stores values such as **Library, Hostel, Transport, Medical Centre, and Wi-Fi**. Whenever a student mentions any of these services, Dialogflow automatically identifies the entity and returns the relevant information.

Using entities improves the chatbot's flexibility and allows it to understand different variations of the same query.



@services

SAVE

☒ Define synonyms ☐ Regexp entity ☐ Allow automated expansion
☐ Fuzzy matching 

 Separate synonyms by pressing the **enter**, **tab** or **;** key.

Library	Library	
Hostel	Hostel	 
Transport	Transport	
Medical	Medical	
WiFi	WiFi	Enter synonym

Conversation Flow

The conversation between the student and the chatbot follows a structured process.

1. Student enters a query.
2. Dialogflow receives the query.
3. Natural Language Processing analyzes the sentence.
4. Dialogflow identifies the appropriate intent.
5. Relevant entities are extracted.
6. The chatbot generates the appropriate response.
7. If no matching intent is found, the Default Fallback Intent is activated.

Search Strategy Used

The Student Support Assistant uses a search strategy similar to Breadth First Search (BFS) for intent selection. When a user enters a query, Dialogflow compares it with all available intents at the same conversational level before selecting the most appropriate response.

Breadth First Search is suitable because it explores all possible intent matches systematically and chooses the best matching intent based on the training phrases.

Advantages

- Fast intent selection
- High response accuracy
- Simple implementation
- Suitable for educational chatbots

Agent	
USER SAYS	COPY CURL
hi	
 DEFAULT RESPONSE ▼	
Greetings! How can I assist?	
INTENT	
Default Welcome Intent	
ACTION	
input.welcome	

Agent	
USER SAYS	COPY CURL
library timings	
 DEFAULT RESPONSE ▼	
Campus services include Library, Hostel, Transport, Medical Centre and Student Support Office.	
INTENT	
Campus Services	
ACTION	
Not available	

USER SAYS

[COPY CURL](#)

how do i register?



DEFAULT RESPONSE



Course registration opens at the beginning of every semester. Please login to the Student Portal and complete the registration before the deadline.

INTENT

[Course Registration](#)

Agent

USER SAYS

[COPY CURL](#)

attendance



DEFAULT RESPONSE



You can view your attendance percentage through the Student Portal.

INTENT

[Attendance](#)

ACTION

Not available

USER SAYS

[COPY CURL](#)

Who won the IPL?



DEFAULT RESPONSE



Say that one more time?

CONTEXTS

[RESET CONTEXTS](#)

INTENT

[Default Fallback Intent](#)

ACTION

input.unknown

USER SAYS

[COPY CURL](#)

show timetable



DEFAULT RESPONSE



Your timetable can be accessed through the Academic Portal.

INTENT

[Timetable](#)

ACTION

Not available

PARAMETER

VALUE

date-time

USER SAYS

[COPY CURL](#)

exam schedule



DEFAULT RESPONSE



The examination schedule is available on the university website under the Examination section.

INTENT

[Exam Schedule](#)

ACTION

Not available

USER SAYS

[COPY CURL](#)

course registration



DEFAULT RESPONSE



Course registration opens at the beginning of every semester. Please login to the Student Portal and complete the registration before the deadline.

INTENT

[Course Registration](#)

USER SAYS

[COPY CURL](#)

exam timetable



DEFAULT RESPONSE



The examination schedule is available on the university website under the Examination section.

INTENT

[Exam Schedule](#)

USER SAYS

[COPY CURL](#)

attendance



DEFAULT RESPONSE



You can view your attendance percentage through the Student Portal.

INTENT

[Attendance](#)

USER SAYS

[COPY CURL](#)

library timings



DEFAULT RESPONSE



Campus services include Library, Hostel, Transport, Medical Centre and Student Support Office.

INTENT

[Campus Services](#)

USER SAYS

[COPY CURL](#)

transport



DEFAULT RESPONSE



Campus services include Library, Hostel, Transport, Medical Centre and Student Support Office.

INTENT

[Campus Services](#)

Evaluation

The chatbot successfully responded to all predefined student queries with high accuracy. Intent recognition was performed correctly for course registration, examination schedule, attendance, timetable, and campus services. Unknown queries activated the Default Fallback Intent, preventing incorrect responses.

Overall, the chatbot demonstrated reliable performance, quick response time, and effective handling of student support requests.

Future Improvements

Future enhancements can make the chatbot more intelligent and user-friendly. The chatbot can be integrated with the university database to provide personalized attendance and examination details. Voice-based interaction can be added using speech recognition. Multilingual support can help students communicate in regional languages. The chatbot can also be connected to mobile applications and Generative AI technologies to provide more detailed and conversational responses.

Conclusion

The AI-based Student Support Assistant successfully demonstrates how Artificial Intelligence can improve university support services. By using Dialogflow, Natural Language Processing, intents, entities, and conversational design, the chatbot provides quick, accurate, and user-friendly responses to common student queries. The system reduces administrative workload, improves response efficiency, and enhances the overall student experience. With additional features such as database integration, multilingual support, and voice interaction, the chatbot can become a comprehensive digital assistant for educational institutions.